

$$t \in [-\infty; +\infty). \quad \xrightarrow{\quad} \quad \vec{r} = \vec{r}_0 + \alpha t$$

t — параметр

$$\vec{r}(t), \quad t \in \mathcal{D}$$

$$t \rightarrow t_0 \quad |\vec{r}(t) - \vec{r}_0| \xrightarrow[t \rightarrow t_0]{} 0$$

$$\text{мы } \exists \vec{r}_1(t), \vec{r}_2(t)$$

$$\lim_{t \rightarrow t_0} \vec{r}_i(t)$$

$$1) \quad \lim_{t \rightarrow t_0} (\vec{r}_1(t) \pm \vec{r}_2(t)) = \vec{r}_1' \pm \vec{r}_2'$$

$$2) \quad \lim_{t \rightarrow t_0} \mathcal{M}_i(\vec{r}(t)) = \mathcal{M}_i \vec{r}_0'$$

$$3) \quad \lim_{t \rightarrow t_0} \vec{r}_1(t) \cdot \vec{r}_2(t) = \vec{r}_1' \cdot \vec{r}_2'$$

$$|[\vec{r}_1(t) \vec{r}_2(t)] - [\vec{r}_1' \vec{r}_2']| \xrightarrow[t \rightarrow t_0]{} 0 \quad (\text{?})$$

$$|[\bar{r}_1(t), \bar{r}_2(t)] - [\bar{r}_1(t), \bar{r}_0^2] + [\bar{r}_1(t), \bar{r}_0^2] - [\bar{r}_0^1, \bar{r}_0^2]| =$$

$$= |[\bar{r}_1(t), \bar{r}_2(t) - \bar{r}_0^2] + \underbrace{[\bar{r}_1(t) - \bar{r}_0^1, \bar{r}_0^2]}_{\rightarrow 0}| =$$

$$|[\bar{r}_1(t), \bar{r}_2(t) - \bar{r}_0^2]|$$

$\rightarrow 0$

$$= |[\bar{r}_0^1, (\bar{r}_2(t) - \bar{r}_0^2)] + [\bar{r}_0^1, \bar{r}_2(t) - \bar{r}_0^2] + [\bar{r}_1(t) - \bar{r}_0^1, \bar{r}_0^2]| \xrightarrow{t \rightarrow t_0} 0$$

НЕПРЕРЫВНОСТЬ

Опр. $\bar{r}(t)$ вектор-функция **непрерывна** в t_0 , если

$$\exists \lim_{t \rightarrow t_0} \bar{r}(t) = \bar{r}(t_0).$$

Производная

$$\text{Опр.} \quad \lim_{\Delta t \rightarrow 0} \frac{\bar{r}(t + \Delta t) - \bar{r}(t)}{\Delta t} = \bar{r}'(t).$$

— **производная**
вектор-функции

Соблюдая: Возьмем $\bar{r}_1(t)$, $\bar{r}_2(t)$ и $\mu(t)$
↑
скаляр

$$1) (\bar{r}_1(t) \pm \bar{r}_2(t))' = \bar{r}_1'(t) \pm \bar{r}_2'(t).$$

$$2) (\bar{r}_1(t) \cdot \mu(t))' = \mu'(t) \bar{r}_1(t) + \bar{r}_1'(t) \mu(t).$$

$$3) (\bar{r}_1(t) \bar{r}_2(t))' = \bar{r}_1'(t) \bar{r}_2(t) + \bar{r}_1(t) \bar{r}_2'(t).$$

$$4) [\bar{r}_1(t) \bar{r}_2(t)]' = [\bar{r}_1'(t), \bar{r}_2(t)] + [\bar{r}_1(t), \bar{r}_2'(t)]$$

$$= \lim_{\Delta t \rightarrow 0} \frac{[\bar{r}_1(t+\Delta t), \bar{r}_2(t+\Delta t)] - [\bar{r}_1(t), \bar{r}_2(t)]}{\Delta t} =$$

$$\lim_{\Delta t \rightarrow 0} \frac{[\bar{r}_1(t+\Delta t), \bar{r}_2(t+\Delta t)] - [\bar{r}_1(t), \bar{r}_2(t+\Delta t)] +$$

$$+ [\bar{r}_1(t), \bar{r}_2(t+\Delta t)] - [\bar{r}_1(t), \bar{r}_2(t)]}{\Delta t} =$$

$$\lim_{\Delta t \rightarrow 0} \frac{[\bar{r}_1(t+\Delta t) - \bar{r}_1(t), \bar{r}_2(t+\Delta t)] + [\bar{r}_1(t), \bar{r}_2(t+\Delta t) - \bar{r}_2(t)]}{\Delta t} =$$

$$= [\bar{r}_1'(t), \bar{r}_2(t)] + [\bar{r}_1(t), \bar{r}_2'(t)].$$

$$\bar{r}(t) \in V^3 \Rightarrow \bar{r}(t) = x(t)\bar{e}_1 + y(t)\bar{e}_2 + z(t)\bar{e}_3.$$

↑
в пространстве

формула Тейлора

Вектор-функции можно считать $\bar{r}^{(k)}(t)$.

$$\bar{r}(t+\Delta t) = \bar{r}(t) + \bar{r}'(t) \Delta t + \bar{r}''(t) \frac{(\Delta t)^2}{2} + \dots$$

1!

2!

$$+ r^{(k)}(t) \frac{(\Delta t)^k}{k!} + \vec{\epsilon}(t, \Delta t).$$

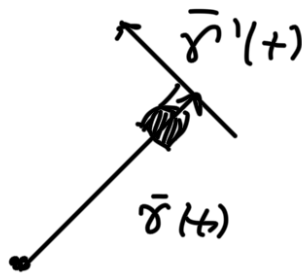
$$\lim_{\Delta t \rightarrow 0} \frac{\vec{\epsilon}(t, \Delta t)}{(\Delta t)^k} = \vec{0}.$$

$$\text{Пусть } |\vec{r}(t)| = \text{const}$$

$$(\vec{r}'(t), \vec{r}(t)) = 0$$

$$|\vec{r}(t)|^2 = (\vec{r}(t), \vec{r}(t))$$

||
const



$$0 = (\vec{r}(t), \vec{r}(t))' = \vec{r}(\vec{r}'(t), \vec{r}(t)). \rightarrow \text{производная ортогональна функции}$$

$$\vec{r}(t) = \mu(t) \cdot \vec{e}$$

прямолин. движение

$$\vec{r}'(t) = \underbrace{\mu'(t)\vec{e}}_{\uparrow} + \mu(t)\vec{e}' = 0$$



производная коллинеарна
линии движения

Если производная представлена в таком виде,
то ~~функция~~ коллинеарна „производной“ и
вектору vice versa

однонапр-е

$$\bar{r}(t) = \mu(t) \cdot \bar{e}(t) \quad (r'(t) = \mu'(t) \bar{e}(t) + \mu(t) \bar{e}'(t))$$

$$\mu(t) = |\bar{r}(t)|.$$

$(\bar{e}(t), \bar{e}'(t)) = 0$; произвольная — вектор-функция

Опр. Пусть есть $\bar{r}'(t)$ и $\bar{g}(t)$: $\bar{g}'(t) = \bar{r}(t)$
 $\bar{g}(t)$ — неопределённый интеграл
 $\bar{g}(t) = \int \bar{r}(t) dt$.

Опр. $\int_a^b \bar{r}(t) dt = \bar{g}(b) - \bar{g}(a)$.

— Неопределённых интегралов скалярно
 и векторно

Свойства

1) $\bar{r}(t) = \lambda r_1(t)$

$$\int_a^b \bar{r}(x) dx = \lambda \int_a^b r_1(x) dx.$$

$$2) \int_a^b (\bar{r}_1(x) \pm \bar{r}_2(x)) dx = \int_a^b \bar{r}_1(x) dx \pm \int_a^b \bar{r}_2(x) dx.$$

$$3) \frac{d}{dt} \int_a^t \bar{r}(x) dx = \bar{r}(t).$$

" " $\int_a^b \bar{r}(x) dx = \bar{r}(b) - \bar{r}(a)$ — скалярно

$$4) \int_a^b \gamma(\tau) d\tau = \int_a^c \gamma(\tau) d\tau + \int_c^b \gamma(\tau) d\tau.$$

$$5) \int_a^b (\bar{\varphi}, \gamma(\tau)) d\tau = (\bar{\varphi}, \int_a^b \gamma(\tau) d\tau)$$

Векторные функции нескольких аргументов

Опр. $\bar{\gamma}(x_1, x_2, \dots, x_n)$ — в.ф. многих аргументов

Опр.

$$\lim_{\Delta x_i \rightarrow 0} \frac{\bar{\gamma}(x_1, \dots, \underbrace{x_i + \Delta x_i}_0, \dots, x_n) - \bar{\gamma}(x_1, \dots, x_i, \dots, x_n)}{\Delta x_i}$$

$$= \frac{\partial \bar{\gamma}}{\partial x_i}.$$

— частная производная в-ф по i -му аргументу

Опр. в-ф двух арг-ов $\bar{\gamma}(u, v)$

есть 2 пр-е: $\frac{\partial \bar{\gamma}}{\partial u}, \frac{\partial \bar{\gamma}}{\partial v}$.

$\bar{\gamma}(u, v)$ дифференцируема, если

$$\exists \lim \bar{\gamma}(u + \Delta u, v + \Delta v) - \bar{\gamma}(u, v) =$$

$$= \underbrace{\frac{\partial \bar{\gamma}}{\partial u} \cdot \Delta u + \frac{\partial \bar{\gamma}}{\partial v} \cdot \Delta v}_{\text{}} + \bar{E}(u, v, \Delta u, \Delta v),$$

$$\frac{\vec{E}}{\sqrt{\Delta u^2 + \Delta v^2}} \xrightarrow[\Delta v \rightarrow 0]{\Delta u \rightarrow 0} 0$$

Дифференциал